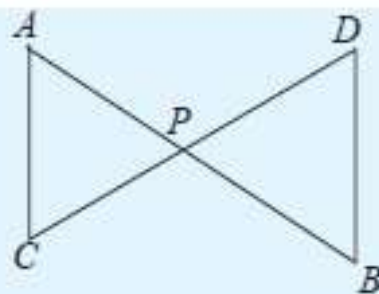
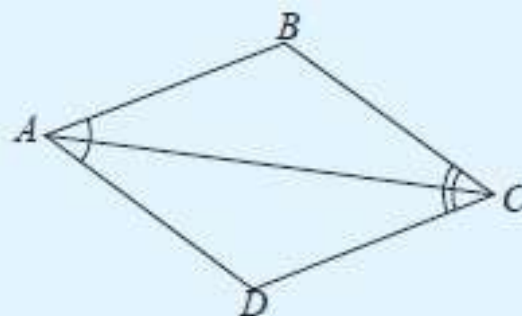


Exercise 8.4

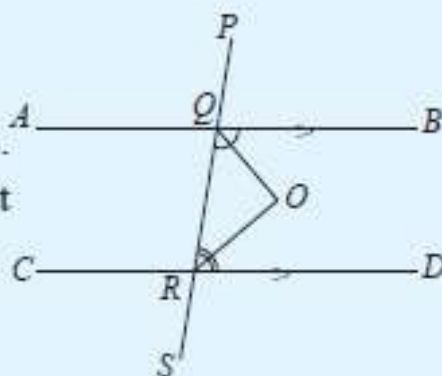
1. In the given figure, $\angle ACP = \angle PBD$. Prove that $\angle CAP = \angle PDB$.



2. In the given figure, the diagonal AC of the quadrilateral $ABCD$ bisects the angles $\angle BAD$ and $\angle BCD$. Prove that $\angle ABC = \angle ADC$.



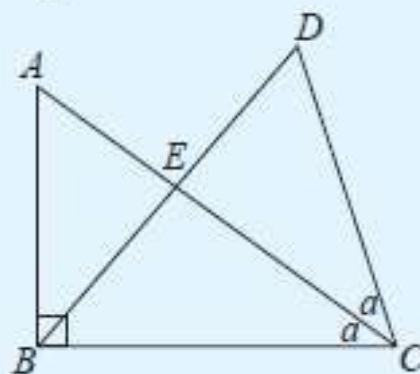
3. In the given figure, AB and CD are parallel lines. The bisectors of the angles $\angle BQR$ and $\angle QRD$ meet at O .



- (i) Find the value of $\angle OQR + \angle QRO$.
 (ii) Prove that $\triangle OQR$ is a right angled triangle.

4. Using the information in the figure

- (i) write down the magnitude of $\angle BAE$, in terms of a .
 (ii) write down the magnitude of $\angle BDC + \angle DBC$ in terms of a .
 (iii) Show that $\angle BDC + \angle DBC = 2 \angle BAE$.



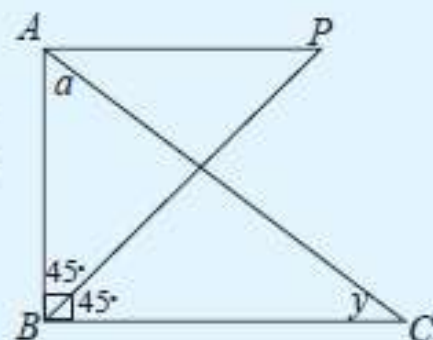
5. In the triangle ABC , $\angle A = \angle B = \angle C$. The bisector of $\angle BAC$ meets the side CB at D .

- (i) Find the magnitude of $\angle BAC$.
 (ii) Prove that $\triangle ABD$ is a right angled triangle.

Miscellaneous Exercise

1. If $\hat{A} + \hat{B} = 110^\circ$, and $\hat{B} + \hat{C} = 120^\circ$ in the triangle ABC , find the magnitude of each angle of the triangle.
2. The magnitude of a $\hat{BAC}$ of the triangle ABC is 100° . The bisectors of the interior angles $\hat{ABC}$ and $\hat{ACB}$ meet at O . Find the magnitude of $\hat{BOC}$.

3. In the figure, the straight line drawn from the point A perpendicular to the side BA of the triangle ABC meets the bisector of $\hat{ABC}$ at P . Prove that $\hat{BAC} + \hat{ACB} = 2 \hat{APB}$.



4. $\hat{ACB} = 3 \hat{ABC}$ in the triangle ABC . The bisector of $\hat{BAC}$ meets BC at E . The point D lies on AE produced such that $AD \perp BD$. Prove that BC is the bisector of $\hat{ABD}$ (Hint: Take $\hat{ABC} = x$ and $\hat{BAC} = 2a$).
5. The line PQ has been drawn through the point A , parallel to the side BC of the triangle ABC . Prove that the sum of the interior angles of the triangle ABC is 180° .